

TOLEDO EXPRESS, OH, USA

WMO: 725360

Lat: 41.587N

Lon: 83.806W

Elev: 205

StdP: 98.88

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94830

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.5	-21.7	0.6	-17.0	-18.7	0.7	-13.5	12.7	-3.7	11.6	-5.4	3.8	240	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.9	23.4	31.4	22.4	29.9	21.6	25.0	30.3	24.0	28.9	23.1	27.7	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.6	27.8	22.5	17.6	26.7	21.7	16.7	25.7	76.7	30.2	73.1	29.0	69.5	27.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.3	8.3	-20.5	35.2	3.7	1.7	-23.2	36.5	-25.4	37.5	-27.5	38.5	-30.2	39.8		
			-20.8	26.5	3.5	1.1	-23.3	27.3	-25.4	27.9	-27.4	28.5	-29.9	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-3.6	-2.1	3.1	9.7	15.8	21.2	23.1	22.1	18.5	11.9	5.2	-0.5
	DBStd	10.63	6.60	6.13	5.96	5.09	4.69	3.72	3.09	2.92	4.09	4.80	5.11	5.62
	HDD10.0	1584	423	340	229	65	6	0	0	0	1	34	160	328
	HDD18.3	3349	680	572	474	263	107	15	2	4	48	207	394	583
	CDD10.0	1742	1	2	14	56	186	336	406	376	254	92	16	3
	CDD18.3	466	0	0	1	4	30	101	149	122	52	8	0	0
	CDH23.3	4197	0	0	7	50	318	926	1408	970	447	69	1	0
Wind	WSAvg	3.7	4.6	4.4	4.3	4.5	3.8	3.2	2.9	2.6	2.8	3.5	4.0	4.2
	PrecAvg	870	57	55	61	84	92	90	84	82	73	66	64	61
	PrecMax	1171	105	136	130	165	163	155	207	224	139	166	181	135
	PrecMin	706	17	13	18	25	33	23	25	19	26	7	23	14
	PrecStd	125	24	32	28	32	42	36	40	45	36	36	32	27
	DB	14.9	17.2	23.9	28.0	31.7	34.5	35.2	33.4	32.9	28.7	21.4	16.6	
	MCWB	12.7	12.1	16.5	18.1	20.7	23.4	24.7	24.4	23.1	20.7	14.1	14.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	11.7	12.8	19.4	24.5	29.3	32.4	33.0	31.4	30.3	25.1	18.0	13.1	
	2% MCWB	10.2	10.3	13.7	15.7	19.7	22.4	23.8	23.2	21.3	17.8	13.8	10.9	
	5% DB	8.5	9.2	15.9	21.8	27.1	30.4	31.3	29.7	28.0	22.6	15.9	10.1	
	5% MCWB	6.6	6.4	11.4	14.1	19.0	21.5	22.8	22.2	20.4	16.7	12.3	7.7	
	10% DB	5.0	5.8	12.4	18.8	24.5	28.5	29.6	28.1	25.9	20.1	13.4	6.9	
	10% MCWB	3.4	3.5	8.8	12.5	17.8	20.7	21.9	21.4	19.3	15.6	10.1	5.1	
	WB	13.3	14.0	17.0	19.5	23.0	25.0	26.6	26.0	24.7	21.9	16.7	14.5	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	14.4	16.1	23.1	26.0	28.6	31.4	32.6	31.8	30.4	26.6	19.2	16.3	
	2% WB	10.5	10.3	14.4	17.2	21.3	23.8	25.1	24.8	23.0	19.7	14.8	11.4	
	2% MCDB	11.4	12.4	18.1	22.6	26.9	30.0	30.7	29.2	27.4	23.3	17.3	12.9	
	5% WB	6.9	6.6	12.1	15.2	20.1	22.9	24.1	23.8	21.8	17.9	12.6	7.9	
	5% MCDB	8.1	9.0	15.5	20.1	25.0	28.5	29.1	27.5	25.7	21.2	15.3	9.6	
	10% WB	3.4	3.8	9.0	13.3	18.7	21.9	23.1	22.7	20.6	16.1	10.4	5.3	
	10% MCDB	4.8	5.7	12.4	18.1	23.2	26.5	27.7	26.4	24.2	19.5	13.1	7.0	
Mean Daily Temperature Range	MDBR	7.5	8.1	9.6	11.1	11.4	11.2	11.3	10.7	11.6	10.6	8.5	6.8	
	5% DB	10.1	11.8	14.6	15.3	14.5	13.5	12.6	12.1	14.0	14.0	11.9	9.6	
	MCWBR	8.9	9.6	9.7	8.7	6.9	5.7	5.3	5.4	6.4	7.6	8.6	8.2	
	5% WB	9.7	11.3	13.4	13.3	12.6	11.6	11.1	10.1	11.4	11.8	10.7	9.0	
	MCWBR	9.0	9.8	9.8	8.7	6.9	5.7	5.3	5.1	5.9	7.3	8.8	8.3	
Clear-Sky Solar Irradiance	taub	0.298	0.334	0.358	0.404	0.433	0.450	0.442	0.434	0.387	0.359	0.324	0.309	
	taud	2.408	2.294	2.317	2.208	2.180	2.176	2.214	2.223	2.338	2.397	2.453	2.404	
	Ebn at Noon	853	866	886	865	846	830	832	827	845	828	815	803	
	Edh at Noon	79	104	115	138	145	146	139	134	112	94	76	74	
All-Sky Solar Radiation	RadAvg	1.54	2.46	3.50	4.63	5.41	6.04	6.07	5.32	4.34	2.86	1.83	1.27	
	RadStd	0.17	0.33	0.31	0.40	0.47	0.41	0.34	0.21	0.35	0.24	0.25	0.13	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	N/A

Nomenclature: See separate page